

MOHAMMED UMAR ANSARI

+91 99207 10332 | umar.231405.it@mhssce.ac.in | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Information Technology undergraduate with hands-on experience in software engineering, backend systems, and cloud-integrated application development. Currently working as a Software Engineer Intern at Bitkraft Technologies LLP, contributing to workbook-based accounting systems, reconciliation engines, document parsing pipelines, and collaborative workflow platforms using React, Node.js, Python, and AWS services. Experienced in developing scalable data-processing systems, cloud-native applications, and automated workflow solutions for accounting and audit use cases. Lead author and co-author of research papers accepted at ICSSSD 2026 focused on agentic AI systems, productivity analytics, and AI-assisted sustainability verification.

EXPERIENCE

Software Engineer Intern

Bitkraft Technologies LLP [23rd Feb, 2026 - Present]

Financial Workflow & Reconciliation Systems

- Designed and developed a **workbook-based accounting workflow platform** supporting independent creation, archival, deletion, and collaborative sharing capabilities.
- Built the application from scratch using **Next.js, Node.js, Python, and Tailwind CSS** based on functional requirements and workflow specifications.
- Implemented **OTP-based authentication workflows** using SMTP server integration and organizational email infrastructure.
- Developed **parsing and normalization pipelines** for accounting vouchers, bank statements, loan statements, and credit card statements across **PDF, CSV, XLSX, and multiple financial document formats**.
- Built an automated **reconciliation and suspense matching engine** with confidence scoring, recommendation-based matching, manual override capabilities, and intelligent status classification workflows.
- Developed **collaborative workbook-sharing functionality** enabling multiple accountants and CA professionals to work simultaneously on shared reconciliation workflows.
- Enabled **export workflows** for reconciliation records and audit-related financial reports.

Cloud & Application Infrastructure

- Developed and deployed a **Single Page Application (SPA)** for fleet fuel leakage estimation and reporting workflows.
- Deployed frontend infrastructure using **AWS S3 Static Hosting** and resolved **CORS integration challenges**.
- Implemented **AWS Lambda-based backend workflows** using Python for automated report generation and email delivery systems.
- Configured **SMTP-based communication workflows** for secure customer communication and lead tracking systems.

AI & Data Pipeline Exposure

- Contributed to **diffusion model experimentation workflows** involving automated dataset collection and preprocessing pipelines.
- Built automated **web and YouTube scraping pipelines** for collecting video datasets used in AI experimentation workflows.
- Collaborated using **Git-based version control** and standard software development workflows within a team environment.

RESEARCH PUBLICATIONS

Cognitrace: An Effort Intelligence System for Real-Time Cross-Context Productivity Analysis

Lead Author | Accepted at ICSSSD 2026 (Paper ID: 319)

- Proposed an **agentic effort intelligence framework** for analyzing productivity patterns using **real-time behavioral analytics** and **cross-context activity analysis**.
- Conducted research focused on **adaptive productivity monitoring**, contextual behavioral interpretation, and intelligent effort evaluation systems.
- Awarded the **Best Paper Award** at ICSSSD 2026.

Green Passport: An Agentic AI-Verified Digital Product Passport for Greenwashing Mitigation

Co-author | Accepted at ICSSSD 2026 (Paper ID: 196)

- Proposed a **digital product passport framework** integrating **AI-assisted sustainability claim verification**, semantic analysis, and cryptographic audit mechanisms.
- Contributed to research focused on **AI-driven sustainability verification** and transparent environmental claim validation systems.

PROJECTS

Ascend APEX - Academic Platform for Excellence (Winner - National Hackathon)

- Built a full-stack academic platform using **Oracle APEX**, **OCI Generative AI**, and **RAG-based retrieval systems**.
- Integrated external APIs and scalable OCI workflows for AI-assisted academic and job-related interactions.
- Awarded **₹100,000** for winning the national-level Oracle APEX Hackathon.

Microservices and CI/CD Pipeline Builder

- Migrated a monolithic application into **Docker-based microservices** with automated **CI/CD pipelines** using AWS services.
- Implemented **blue-green deployments** using **Amazon ECR**, **ECS Fargate**, and **Application Load Balancer** workflows.

Cloud Web Application Builder

- Built a highly available cloud web application using **AWS EC2**, **RDS**, **VPC**, **Auto Scaling**, and **Load Balancer** services.
- Managed secure credential handling using **AWS Secrets Manager** and documented deployment workflows.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C/C++, JavaScript, SQL
- **Backend & Web Development:** React.js, Node.js, Express.js, HTML, CSS, Tailwind CSS, REST APIs, Authentication Systems
- **Data Engineering & Processing:** Data Analysis, Workflow Automation, Financial Data Processing, Report Generation Systems, Document Parsing Pipelines
- **AI/Generative AI:** Generative AI, RAG (Retrieval-Augmented Generation), Prompt Engineering, AI-based Recommendation Systems
- **Cloud & Distributed Systems:** Cloud & Distributed Systems: AWS, OCI, EC2, S3, ECS, ECR, Lambda, RDS, VPC, Docker, CI/CD, Microservices, Serverless Architecture
- **Developer Tools:** Git, GitHub, Kiro, OpenAI Codex, Antigravity, GitHub Copilot

CERTIFICATIONS

- **Amazon Web Services (x2)** - AWS Certified Cloud Practitioner (CLF-C02) and AWS Certified AI Practitioner (AIF-C01)
- **Oracle Cloud Certifications (x6):** OCI Foundations, AI Foundations, Data Platform Foundations, Generative AI Professional, AI Vector Search Professional, APEX Developer Professional
- **Red Hat Academy (x3):** Python Programming with Red Hat (AD141 - RHA), Red Hat Enterprise Linux Automation with Ansible (RH294) and Red Hat System Administration I (RH124 - RHA)
- **CodeSignal:** Prompt Engineering for Everyone, Mastering Cloud Engineering with AWS & Python, Building Generative AI Applications with AWS, Enterprise Machine Learning with Amazon SageMaker

LEADERSHIP & ACHIEVEMENTS

- Treasurer, **ACM MHSSCOE Student Chapter (2026-27).**
- Director of Operations, **AWS Student Builder Group MHSSCOE (2026-27).**
- **AWS Student Builder Campus Leader (AWS SBCL) 2026**
- Winner, **National Oracle APEX Hackathon Xcelerate25** – ₹100,000 prize

EDUCATION

Bachelor of Engineering in Information Technology MH Saboo Siddik College of Engineering, Mumbai Expected Graduation: May 2027 CGPA: 8.05	2023 - 2027 MUMBAI UNIVERSITY
--	--